BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 044 · Evaluating Technology Impacts

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is a key feature of an open-source licence?

Q2. How does the OS support multitasking?
A. Adds a CPU per app
B. Closes every other app
C. Gives each app all RAM
D. Shares CPU time

Q3. “A longer recording automatically has finer amplitude accuracy.” Correct this. Explain why.

Q4. A mono clip uses 1,000 samples/s for 3 s at 8 bits/sample. Calculate its raw size in bytes, excluding headers and compression.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Evaluate — Weigh strengths and weaknesses and reach a supported judgement.

Stakeholder — Person or group affected by a decision.

Justification — Reasons and evidence supporting a choice.

Privacy — Control and protection of information about individuals.

Ethical impact — Effect on fairness and responsible behaviour.

Environmental impact — Effect on energy, resources, pollution or waste.

New learning

Evaluation is not a list: Describe the proposed technology and its context first. A strong evaluation considers different viewpoints, consequences and realistic trade-offs.

Think about stakeholders: Users, employees, managers and the wider community may be affected differently. Give real impacts rather than simply saying technology is good or bad.

Consider multiple dimensions: Discuss appropriate ethical, legal, environmental, cultural or privacy aspects. Do not force every category into every answer if it is irrelevant.

Reach a clear judgement: Make a reasoned decision that recognises the most important benefit and risk in the given scenario. Explain conditions that would make the decision more acceptable.

Worked example

A school plans to use automated facial recognition to record attendance. Evaluate TWO benefits and TWO concerns, then give a supported judgement. [6 marks]

1. Identify the aim: recording attendance quickly and accurately.

2. Automation may reduce administrative time [1] so staff can focus on teaching [1].

3. Face data is sensitive personal information and may produce privacy or fairness concerns [1]; inaccurate recognition could affect pupils [1].

4. A justified judgement [2]: only consider use with lawful safeguards, accuracy checks and a fair non-biometric alternative. Credit reasoned alternative conclusions.

We Do · Practise with guidance

Guided 1

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

A town council plans to replace most paper communications with a mobile app. Evaluate benefits and drawbacks, with a judgement. [6 marks]

Worked method: An app may speed communication [1] and reduce paper/postage [1]. People without smartphones/digital skills could be excluded [1], harming fair access [1]. A reasoned judgement [2] could support the app if telephone and paper alternatives remain available.

Guided 2

A company allows staff to work remotely using cloud services. Evaluate an advantage and a concern. [4 marks]

Worked method: Staff can access work from different locations [1], improving flexibility [1]. Cloud access depends on connection/security safeguards [1], so outages or poor access control could disrupt work or expose data [1].

Guided 3

A school installs more computers rather than repairing current ones. Evaluate an environmental and social consequence. [4 marks]

Worked method: More computers may improve access for students [1], reducing some classroom waiting or sharing [1]. Replacing working machines creates e-waste/material demand [1], increasing environmental impact [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict the advice when a service is online-only.

```
onlineOnly = true
if onlineOnly then
    print("Offer alternatives")
endif
```

predict

Question 1

State TWO groups who could be affected by a new school app. [2 marks] Source: Original OCR J277-style teaching question; not an official OCR past paper.

Question 2

A sports club moves registration entirely online. Explain one benefit and one drawback for members. [4 marks] Source: Original OCR J277-style teaching question; not an official OCR past paper.

Question 3

Evaluate a company's proposal to monitor employee locations with a work app. Consider ethical and privacy effects and reach a judgement. [6 marks] Source: Original OCR J277-style teaching question; not an official OCR past paper.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Exit Ticket · Answer before checking

A strong evaluation identifies relevant benefits and harms and supports its conclusion.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It permits access to source code and modification or redistribution under its conditions.

Q2. D — Shares CPU time

Q3. It doubles size but does not by itself improve sampling quality.

Q4. (a) 1,000 × 3 × 8 = 24,000 bits; divide by 8 = 3,000 bytes.

I Do / We Do pages · Guided practice

Q1. An app may speed communication [1] and reduce paper/postage [1]. People without smartphones/digital skills could be excluded [1], harming fair access [1]. A reasoned judgement [2] could support the app if telephone and paper alternatives remain available.

Q2. Staff can access work from different locations [1], improving flexibility [1]. Cloud access depends on connection/security safeguards [1], so outages or poor access control could disrupt work or expose data [1].

Q3. More computers may improve access for students [1], reducing some classroom waiting or sharing [1]. Replacing working machines creates e-waste/material demand [1], increasing environmental impact [1].

You Do pages · Code First

```
onlineOnly = true
if onlineOnly then
    print("Offer alternatives")
endif
```

Expected result: Offer alternatives

You Do pages · Independent questions

Q1. Pupils [1] and parents/carers [1]. Teachers/administrators also acceptable.

Q2. Members with devices can register remotely [1], saving travel/time [1]. Members without connectivity or digital skills [1] may be unable to register easily [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Q3. Location monitoring may help coordinate work or safety [1], giving managers useful scheduling information [1]. It may intrude into employees' privacy [1] or harm trust [1]. A justified conclusion [2] should weigh necessity, lawfulness, transparency, limits and proportionate alternatives.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: